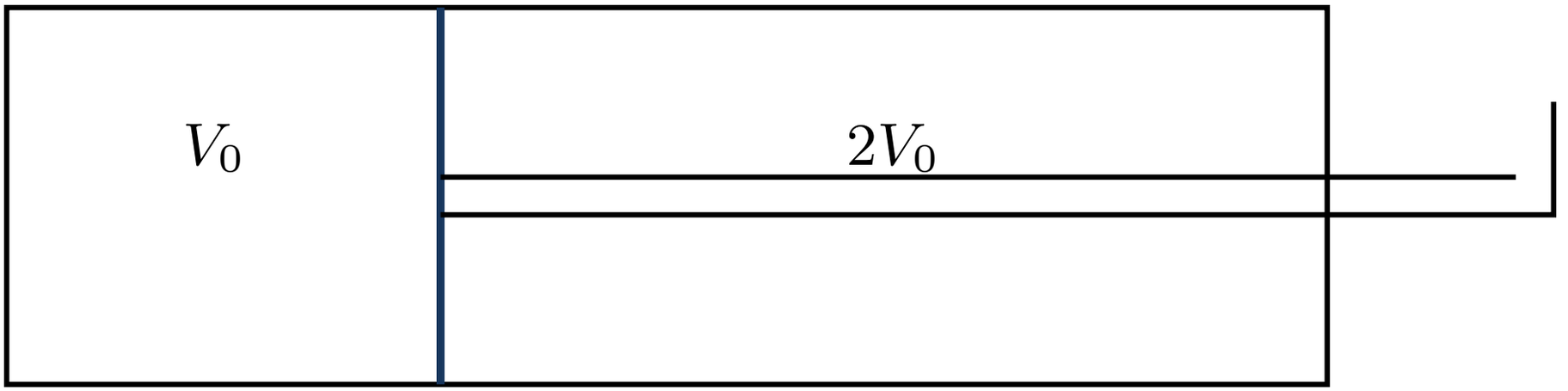
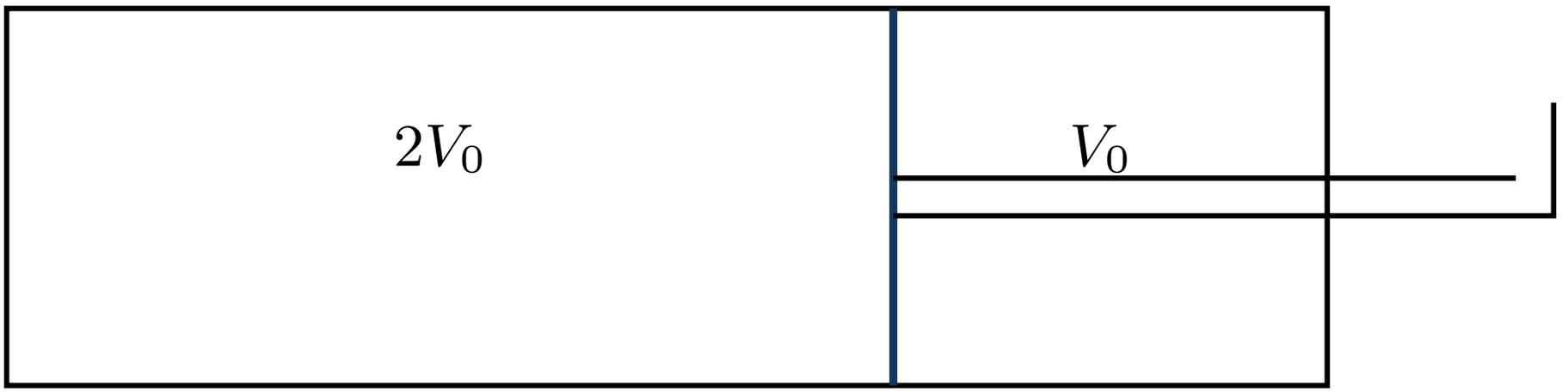
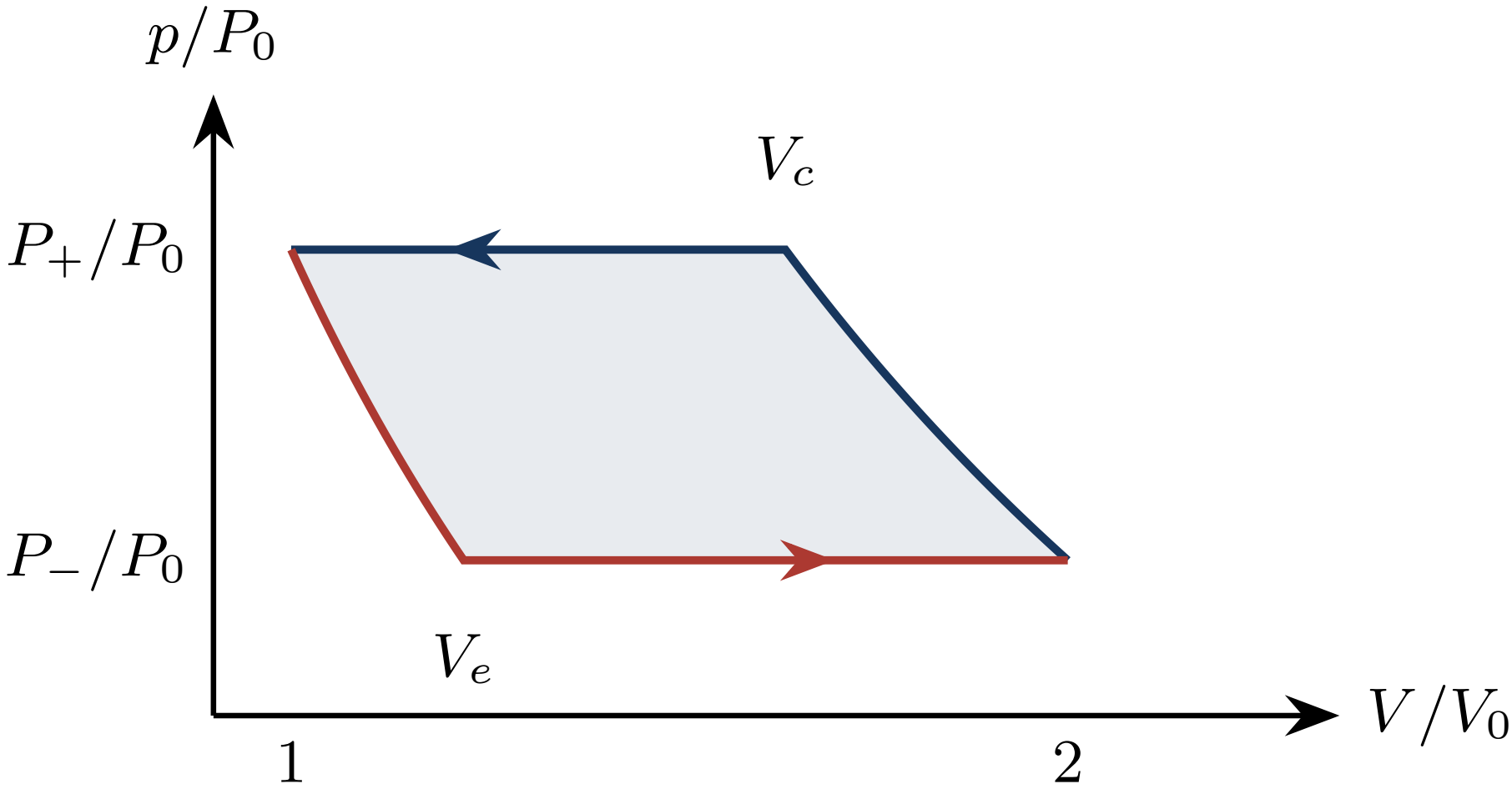
两侧各有进气、出气单向阀



向右推



向左推



图示 $\Delta P/P_0 = 0.1$

压缩：两阀关闭

$pV = 2P_-V_0$

$V = V_c$

出气阀开

$p = P_+$，n 减少

膨胀：两阀关闭

$pV = P_+V_0$

$V = V_e$

进气阀开

$p = P_-$，n 增加

压缩末态与膨胀初态相同，反之亦然

压缩闭阀功

$2P_-V_0\ln(P_+/P_-)$

膨胀闭阀功

$P_+V_0\ln(P_+/P_-)$

压缩开阀功

$(2P_- - P_+)V_0$

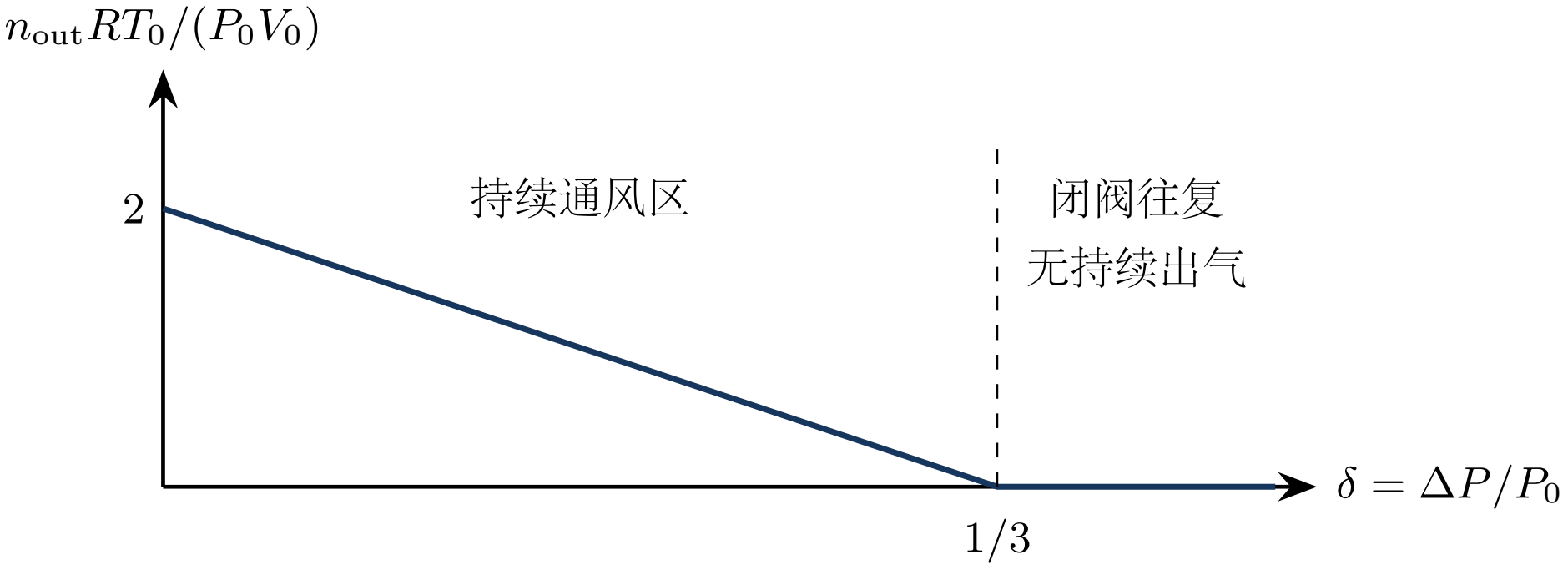
相抵

膨胀开阀功

$(2P_- - P_+)V_0$

单室净功 $(2P_- - P_+)V_0\ln(P_+/P_-)$

两室完整周期再乘 2


$n_{\text{out}}RT_0/(P_0V_0)$
2
持续通风区
闭阀往复
无持续出气
$\delta = \Delta P/P_0$
1/3